26/10/26

Signal Processing

- Recap:- Inverse Systems, All-pass Systems, Min-phase Systems
  - Causal + stable systems (rational system functions)
- Relating magnitude and phase response
  - | |, x, o; ∠, x, o, min-phase
  - $|H(e^{j\omega})|^2$
  - Examples

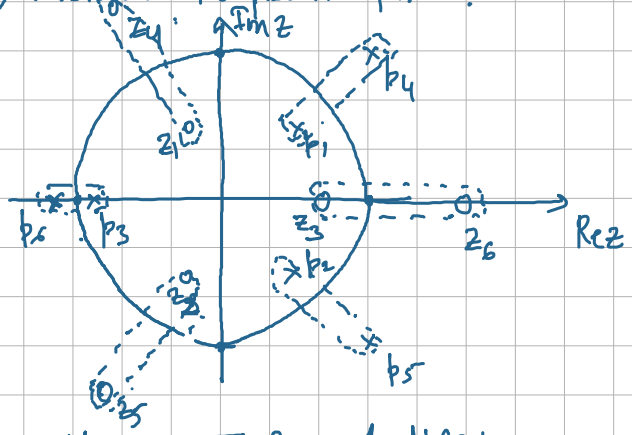
Announcements
- MTR due on 27/10
- Quiz
- Coverage:
  - This week:
    - Magnitude & phase relation
    - Genl lin phase
    - Filter design - IIR, FIR (Next week)
  - last week:
    - Sampling

1: • $H_1(z) = \left[2(1-z^{-1})(1+0.5z^{-1})\right] / \left[(1-0.8e^{j\pi/4}z^{-1})(1-0.8e^{j\pi/4}z^{-1})\right]$

• $H_2(z) = \left[(1-z^{-1})(1+2z^{-1})\right] / \left[(1-0.8e^{j\pi/4}z^{-1})(1-0.8e^{-j\pi/4}z^{-1})\right]$

Show that $C_1(z) = C_1(z)$.

2: C(z) with poles and zeros marked below. If system is stable, causal, and a_k, b_k are real, what are the possible options?



- Recall: $H(z) = \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=1}^{M}(1-c_k z^{-1})}{\prod_{k=1}^{N}(1-d_k z^{-1})}$ (rational system)

  Assuming b_0, a_0 are real.

  ← zeros of H(z)

  ← poles of H(z)

- Q Given H(z), find $H_{inv}(z)$. Can $H_{inv}(z) = 1/H(z)$?
- We analyzed the case where H(z) is causal and stable.
- Causality and stability require the poles of the system $\hat{H}(z)$ to be inside the unit circle.

- Causality and stability of $H_{av}(z)$ impose restrictions on the locations of the zeros of $H(z)$.

$$\Rightarrow \quad \max_k |c_k| < 1$$

- $H_{ap}(z) = \left(\frac{z^{-1} - a^*}{1 - a z^{-1}}\right)$; pole and zero located at conjugate reciprocal locations in the z-plane.

- $H_{min}(z)$: A system whose poles and zeros are inside the unit circle.

- Recap: Any <u>causal and stable system</u> $H(z) = H_{min}(z) \cdot H_{ap}(z)$

- Q. Can any rational LTI system $H(z) = H_{min}(z)\, H_{ap}(z)$? Yes. Same process.

- Recall: $H(z) = \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=1}^{M} (1 - c_k z^{-1})}{\prod_{k=1}^{N} (1 - d_k z^{-1})}$; <u>Causal + Stable</u>

Let c_1 be outside the unit circle.

$$= \frac{b_0}{a_0} \frac{\prod_{k=2}^{M} (1 - c_k z^{-1})}{\prod_{k=1}^{N} (1 - d_k z^{-1})} \cdot \frac{(1 - c_1 z^{-1}) \cdot (z^{-1} - c_1^*)}{(z^{-1} - c_1^*)}$$

$$= \underbrace{\left(\frac{b_0}{a_0}\right) \frac{\prod_{k=2}^{M} (1 - c_k z^{-1}) \cdot (z^{-1} - c_1^*)}{\prod_{k=1}^{N} (1 - d_k z^{-1})}}_{H_{min}(z)} \cdot \underbrace{\left(\frac{1 - c_1 z^{-1}}{z^{-1} - c_1^*}\right)}_{H_{ap}(z)}$$

- $H(z) = \frac{(z-3)}{(z-5)} = \frac{(1-3z^{-1})}{(1-5z^{-1})} = \frac{(1-3z^{-1})}{(1-5z^{-1})} \cdot \frac{(z^{-1}-3)}{(z^{-1}-3)} \cdot \frac{(z^{-1}-5)}{(z^{-1}-5)}$

$$= \underbrace{\left(\frac{z^{-1}-5}{z^{-1}-3}\right)}_{\text{M'in phase}} \underbrace{\left(\frac{1-3z^{-1}}{z^{-1}-3}\right)\left(\frac{z^{-1}-5}{1-5z^{-1}}\right)}_{\text{All pass}}$$

Q. Given a rational LTI system, is there a relation between its magnitude and phase response? $H(z) = \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=1}^{M}(1-c_k z^{-1})}{\prod_{k=1}^{N}(1-d_k z^{-1})}$. In general, no.

Motivation:

A: This question is important since problems in filter design typically specify the magnitude response.

- In the case of causal and stable systems, we can say something useful about the relation b/w mag & phase response.
- Given $|H(e^{j\omega})|$, $=$ causal + stable and specify the number of poles and zeros. Can we say something about $\angle H(e^{j\omega})$?

$$= |H(z)|_{z=e^{j\omega}}$$

- Note that $H_{ap}(z) = A \underbrace{\prod_{k=1}^{N_r} \frac{(z^{-1}-d_k)}{(1-d_k z^{-1})}}_{\text{real}} \cdot \underbrace{\prod_{k=1}^{N_c} \frac{(z^{-1}-e_k^*)}{(1-e_k z^{-1})} \frac{(z^{-1}-e_k)}{(1-e_k^* z^{-1})}}_{\text{complex}}$